Exercise on Earring Selection

- Mary has **9 different earrings**. Mary must wear at least one earring. Each ear has two piercings for wearing an earring. It is not necessary for her to wear all the 4 earrings on her ears, e.g. she can wear one earring and leave the other 3 piercings empty. How many possible ways for Mary to wear her 9 earrings if the position of the earrings matters?
- Solution:
- If Mary wears 4 earrings there are 9x8x7x6 permutations.
- If Mary wears 3 earrings we must select the 3 piercings in C(4,3) ways, and permute the earrings in 9x8x7 ways.
- If Mary wears 2 earrings we must select the 2 piercings in C(4,2) ways, and permute the earrings in 9x8 ways.
- If Mary wears 1 earring we must select the 1 piercing in C(4,1) ways, and permute the earrings in 9 ways.
- Sum of all ways is: C(4,3)x9x8x7 + C(4,2)x9x8 + C(4,1)x9 = 5508